3 Yr. Degree/4 Yr. Honours 4th Semester Examination, 2025 (CCFUP)

Subject : Computer Science

Course: COMP4011 (MAJOR)

Time: 2 Hours **Full Marks: 40**

The figures in the right hand margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any five* questions: 2×5=10

(a) What do you mean by spooling?

(b) Define Starvation.

(c) What is Turnaround Time and Response Time?

(d) What are the differences between pre-emptive and non-preemptive scheduling?

(e) What do you mean by Context Switching?

(f) What is physical and virtual address?

(g) What do you mean by semaphore?

(h) Define File.

2. Answer *any two* questions: 5×2=10

(a) Discuss basic functions of Operating System. 5

(b) Define Process. Discuss life cycle of process with the help of a diagram. 1+4

(c) Define Dead Lock. Discuss four necessary conditions for occurance of deadlock. 1+4

(d) Define Page Fault. Discuss different steps taken by Operating System, when a page fault occurs. 1+4

3. Answer *any two* questions: 10×2=20

(a) Consider the set of 5 processes whose arrival time and brust time are given below:

Process	Arrival time	Brust time	Priority
P_0	0	4	2
P_1	1	3	3
P_2	2	1	4
P_3	3	5	5
P_4	4	2	5

Calculate average waiting time and average turn around time if the CPU scheduling policy is Round Robin with time quantum 2 unit and preemptive priority [Higher number represents higher priority] 5+5

(b) (i) What do you mean by Belady's Anomaly? Explain it with the help of example. 7

(ii) Discuss fixed and variable partition memory allocation. 3

(c) Discuss Contiguous and Indexed file allocation methods with the help of suitable diagrams. 5+5

(d) Write short notes on *(any two)* : 5+5

(i) Authentication

(ii) Time Sharing System

(iii) Process Control Block

(iv) Critical Section